

## 1. Prob 16.2

(a) Given

$$V(r) = -\frac{Ze^2}{r} \quad (1)$$

the circular equation of motion is

$$\frac{Ze^2}{r^2} = m\omega_0^2 r \quad \Rightarrow \quad \omega_0 = \sqrt{\frac{Ze^2}{mr^3}} \quad (2)$$

The energy and angular momentum are

$$E = -\frac{Ze^2}{r} + \frac{1}{2}m\omega_0^2 r^2 = -\frac{Ze^2}{2r} \quad L = m\omega_0 r^2 = \sqrt{mZe^2 r} \quad (3)$$

(16.12) is now written as

$$\frac{dE}{dt} = -\frac{\tau}{m} \left( \frac{dV}{dr} \right)^2 \quad \Rightarrow \quad \left( \frac{Ze^2}{2} \right) \frac{1}{r^2} \frac{dr}{dt} = -\frac{\tau}{m} \left( \frac{Ze^2}{r^2} \right)^2 \quad \Rightarrow \quad \frac{1}{3} \frac{dr^3}{dt} = -\frac{2\tau Ze^2}{m} \quad (4)$$

of which the solution is

$$r^3 = r_0^3 - \frac{6\tau Ze^2}{m} t = r_0^3 - 9Z(c\tau)^3 \frac{t}{\tau} \quad (5)$$

Similarly for the angular momentum,

$$\frac{dL}{dt} = -\frac{\tau}{m} \left( \frac{1}{r} \frac{dV}{dr} \right) L \quad \Rightarrow \quad \frac{1}{2} \sqrt{\frac{mZe^2}{r}} \frac{dr}{dt} = -\frac{\tau}{m} \frac{Ze^2}{r^3} \sqrt{mZe^2 r} \quad \Rightarrow \quad \frac{1}{3} \frac{dr^3}{dt} = -\frac{2\tau Ze^2}{m} \quad (6)$$

which is the same governing differential equation as (4).

(b) With

$$r = \frac{n^2 a_0}{Z} \quad (7)$$

substituted into (5), then taking the time derivative of both sides, we have

$$6n^5 \left( \frac{a_0}{Z} \right)^3 \frac{dn}{dt} = -9Z(c\tau)^3 \frac{1}{\tau} \quad \Rightarrow \quad \frac{dn}{dt} = -\frac{3Z^4}{2n^5} \left( \frac{c\tau}{a_0} \right)^3 \frac{1}{\tau} = -\frac{2}{3} \left( \frac{e^2}{\hbar c} \right) \left( \frac{Ze^2}{\hbar c} \right)^4 \left( \frac{mc^2}{\hbar} \right) \frac{1}{n^5} \quad (8)$$

agreeing with problem 14.21.

(c) From (5), the time to fall from radius  $r_1$  to  $r_2$  is

$$t = \left[ \frac{r_1^3 - r_2^3}{9Z(c\tau)^3} \right] \tau \quad (9)$$

By (7), the time it takes to fall from  $n_1$  to  $n_2$  is

$$t = \frac{\tau(n_1^6 - n_2^6)}{9Z^4} \left( \frac{a_0}{c\tau} \right)^3 = \frac{3\tau(n_1^6 - n_2^6)}{8Z^4 \alpha^6} \quad (10)$$

For a mu meson  $m_\mu = 207m_e$ ,  $\tau_\mu = \tau_e/207 = 6.26 \times 10^{-24}/207 = 3.02 \times 10^{-26}$  s. This gives

$$t_{10 \rightarrow 4} = \frac{3(3.02 \times 10^{-26})(10^6 - 4^6)}{8} \cdot 137^6 \approx 7.5 \times 10^{-8} \text{ s} \quad (11)$$

$$t_{4 \rightarrow 1} = \frac{3(3.02 \times 10^{-26})(4^6 - 1)}{8} \cdot 137^6 \approx 3.1 \times 10^{-10} \text{ s} \quad (12)$$

## 2. Prob 16.3

(a) We use (16.13) to find the decay in binding energy

$$\left\langle \frac{d\epsilon}{dt} \right\rangle = - \left\langle \frac{dE}{dt} \right\rangle = \frac{\tau}{m} \left\langle \left( \frac{dV}{dr} \right)^2 \right\rangle = \frac{\tau Z^2 e^4}{m} \left\langle \frac{1}{r^4} \right\rangle \quad (13)$$

where

$$\left\langle \frac{1}{r^4} \right\rangle = \frac{\int_0^T \frac{dt}{r^4}}{\int_0^T dt} = \frac{\int_0^{2\pi} \frac{d\theta}{r^4} \frac{dt}{d\theta}}{\int_0^{2\pi} \frac{dt}{d\theta} d\theta} \quad (14)$$

With  $dt/d\theta$  given by Kepler's law

$$\frac{dt}{d\theta} = \frac{mr^2}{L} \quad (15)$$

(14) becomes

$$\left\langle \frac{1}{r^4} \right\rangle = \frac{\int_0^{2\pi} \frac{d\theta}{r^2}}{\int_0^{2\pi} r^2 d\theta} \quad (16)$$

With

$$\frac{1}{r} = \frac{Ze^2 m}{L^2} \left[ 1 + \sqrt{1 - \frac{2\epsilon L^2}{Z^2 e^4 m}} \cos(\theta - \theta_0) \right] \quad (17)$$

we see that

$$r_{\min} = \left\{ \frac{Ze^2 m}{L^2} \left[ 1 + \sqrt{1 - \frac{2\epsilon L^2}{Z^2 e^4 m}} \right] \right\}^{-1} \quad (18)$$

$$r_{\max} = \left\{ \frac{Ze^2 m}{L^2} \left[ 1 - \sqrt{1 - \frac{2\epsilon L^2}{Z^2 e^4 m}} \right] \right\}^{-1} \quad (19)$$

so the semi-major axis  $a$  and the semi-minor axis  $b$  are

$$a = \frac{r_{\min} + r_{\max}}{2} = \frac{Ze^2}{2\epsilon} \quad b = \sqrt{r_{\min} r_{\max}} = \frac{L}{\sqrt{2m\epsilon}} \quad (20)$$

giving the denominator in (16) (i.e., twice the area of the ellipse) as

$$\int_0^{2\pi} r^2 d\theta = 2A = 2\pi ab = 2\pi \cdot \frac{Ze^2}{2\epsilon} \cdot \frac{L}{\sqrt{2m\epsilon}} = \frac{\pi Ze^2 L}{\sqrt{2m\epsilon}^{3/2}} \quad (21)$$

On the other hand, the numerator in (16) can be done by taking the square of (17)

$$\int_0^{2\pi} \frac{d\theta}{r^2} = \left( \frac{Ze^2 m}{L^2} \right)^2 \cdot 2\pi \left[ 1 + \frac{1}{2} \left( 1 - \frac{2\epsilon L^2}{Z^2 e^4 m} \right) \right] = \frac{Z^2 e^4 m^2}{L^4} \pi \left( 3 - \frac{2\epsilon L^2}{Z^2 e^4 m} \right) \quad (22)$$

Putting everything together back to (13) gives

$$\left\langle \frac{d\epsilon}{dt} \right\rangle = \frac{2^{3/2}}{3} \frac{Z^3 e^8 \sqrt{m\epsilon}^{3/2}}{L^5 c^3} \left( 3 - \frac{2\epsilon L^2}{Z^2 e^4 m} \right) \quad (23)$$

Similarly, we use (16.16) for the decay in angular momentum

$$\left\langle \frac{dL}{dt} \right\rangle = -\frac{\tau}{m} \left\langle \frac{1}{r} \frac{dV}{dr} \right\rangle L = -\frac{\tau Ze^2}{m} \left\langle \frac{1}{r^3} \right\rangle L \quad (24)$$

where, as before,

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{\int_0^{2\pi} \frac{d\theta}{r^3} \frac{dt}{d\theta}}{\int_0^{2\pi} \frac{dt}{d\theta} d\theta} = \frac{\int_0^{2\pi} \frac{d\theta}{r}}{\int_0^{2\pi} r^2 d\theta} = \frac{\frac{Ze^2m}{L^2} \cdot 2\pi}{\frac{\pi Ze^2L}{\sqrt{2m}\epsilon^{3/2}}} = \frac{2^{3/2}m^{3/2}\epsilon^{3/2}}{L^3} \quad (25)$$

Putting this back to (24) gives

$$\left\langle \frac{dL}{dt} \right\rangle = -\frac{\tau Ze^2}{m} \cdot \frac{2^{3/2}m^{3/2}\epsilon^{3/2}}{L^3} \cdot L = -\frac{2^{5/2}}{3} \frac{Ze^4}{\sqrt{mc^3}} \frac{\epsilon^{3/2}}{L^2} \quad (26)$$

(b) Dividing the two secular rates from part (a) eliminates the time and the common factor  $\epsilon^{3/2}/c^3$ ,

$$\frac{d\epsilon}{dL} = \frac{\langle d\epsilon/dt \rangle}{\langle dL/dt \rangle} = -\frac{1}{2} \frac{Z^2e^4m}{L^3} \left( 3 - \frac{2\epsilon L^2}{Z^2e^4m} \right) = -\frac{3}{2} \frac{Z^2e^4m}{L^3} + \frac{\epsilon}{L} \quad (27)$$

Rewriting (27) gives

$$\frac{d}{dL} \left( \frac{\epsilon}{L} \right) = -\frac{3}{2} \frac{Z^2e^4m}{L^4} \implies \frac{\epsilon}{L} = \frac{1}{2} \frac{Z^2e^4m}{L^3} + C \quad (28)$$

Imposing  $\epsilon = \epsilon_0$  at  $L = L_0$  yields

$$\epsilon(L) = \frac{Z^2e^4m}{2L^2} \left[ 1 - \left( \frac{L}{L_0} \right)^3 \right] + \frac{\epsilon_0}{L_0} L \quad (29)$$

For the eccentricity, notice that

$$\frac{2\epsilon L^2}{Z^2e^4m} = \left[ 1 - \left( \frac{L}{L_0} \right)^3 \right] + \frac{2\epsilon_0 L^3}{Z^2e^4m L_0} \quad (30)$$

The initial eccentricity  $\eta_0$  is given by

$$\eta_0^2 = 1 - \frac{2\epsilon_0 L_0^2}{Z^2e^4m} \quad (31)$$

therefore

$$\eta^2 = 1 - \frac{2\epsilon L^2}{Z^2e^4m} = \left( \frac{L}{L_0} \right)^3 - \frac{2\epsilon_0 L^3}{Z^2e^4m L_0} = \left( \frac{L}{L_0} \right)^3 \left[ 1 - \frac{2\epsilon_0 L_0^2}{Z^2e^4m} \right] = \left( \frac{L}{L_0} \right)^3 \eta_0^2 \implies \eta = \eta_0 \left( \frac{L}{L_0} \right)^{3/2} \quad (32)$$

Since  $L$  decreases through radiation, the eccentricity decreases as  $(L/L_0)^{3/2}$ , and the orbit becomes increasingly circular with time.

(c) For a circular orbit the eccentricity vanishes,

$$\eta = 0 \implies \epsilon = \frac{Z^2e^4m}{2L^2} = \frac{Ze^2}{2 \left( \frac{L^2}{Ze^2m} \right)} \quad (33)$$

This allows us to define

$$r \equiv \frac{L^2}{Ze^2m} \quad (34)$$

and express everything through  $r$ . The angular-momentum rate (26) becomes

$$\left\langle \frac{dL}{dt} \right\rangle = -\frac{2^{5/2}}{3} \frac{Ze^4}{\sqrt{mc^3}} \frac{\epsilon^{3/2}}{L^2} = -\frac{2^{5/2}}{3} \frac{Ze^4}{\sqrt{mc^3}} \frac{(Ze^2/2r)^{3/2}}{Ze^2mr} = -\tau \cdot \frac{Ze^2}{m} \sqrt{\frac{Ze^2m}{r^5}} \quad (35)$$

Substituting the LHS of (35) with the time derivative of (34) gives

$$\frac{1}{3} \frac{dr^3}{dt} = -\frac{2\tau Ze^2}{m} \quad (36)$$

identical to the governing equation (4) of Problem 16.2, so the elliptic result specialized to  $\eta = 0$  reproduces the circular case.